

The Sun, Earth & Moon

This is your guided version. It uses shorter sentences, word banks, and fill-in-the-blank notes to help you learn one step at a time. Read slowly. Fill in every blank. You can do this!



Georgia Standard S6E2 · Supported Reading

WHAT YOU'LL LEARN

- | | |
|---------------------------------|-----------------------------------|
| 01 The Sun, Earth & Moon | 02 Why the Moon Has Phases |
| 03 The Eight Moon Phases | 04 Solar & Lunar Eclipses |
| 05 Why We Have Seasons | 06 Tilt, Not Distance |

Name: _____

Period: _____

Date: _____

2.1 The Sun, Earth & Moon

My Learning Goal

I can name three motions in space and what each one causes.

Three things in space matter most:

- The **Sun** is a star. It gives us light and heat.
- The **Earth** is our planet. It is our home.
- The **Moon** is a rocky ball that orbits Earth.

Each motion below causes something. Fill in the last column using the word bank.

Word Bank

day and night

moon phases

the seasons

Motion	How Long	What It Causes
Earth spins	1 day	_____
Moon orbits Earth	about 1 month	_____
Earth orbits Sun	1 year	_____

2.2 Why the Moon Has Phases

My Learning Goal

I can explain why the Moon looks different on different nights.

The Moon does **not** make its own light.

The Sun lights up **one half** of the Moon.

As the Moon moves around Earth, we see that lit half from different angles.

The shape we see is called a **phase**.

Watch Out

The dark part of the Moon is **not** Earth's shadow. It is just the half the Sun is not lighting up right now.

Circle the correct answer:

- The Moon's light comes from the (Sun / Earth).

- The Moon's real shape (changes / stays the same) as it orbits.

2.3 The Eight Moon Phases

🎯 My Learning Goal

I can name the moon phases in order and tell waxing from waning.

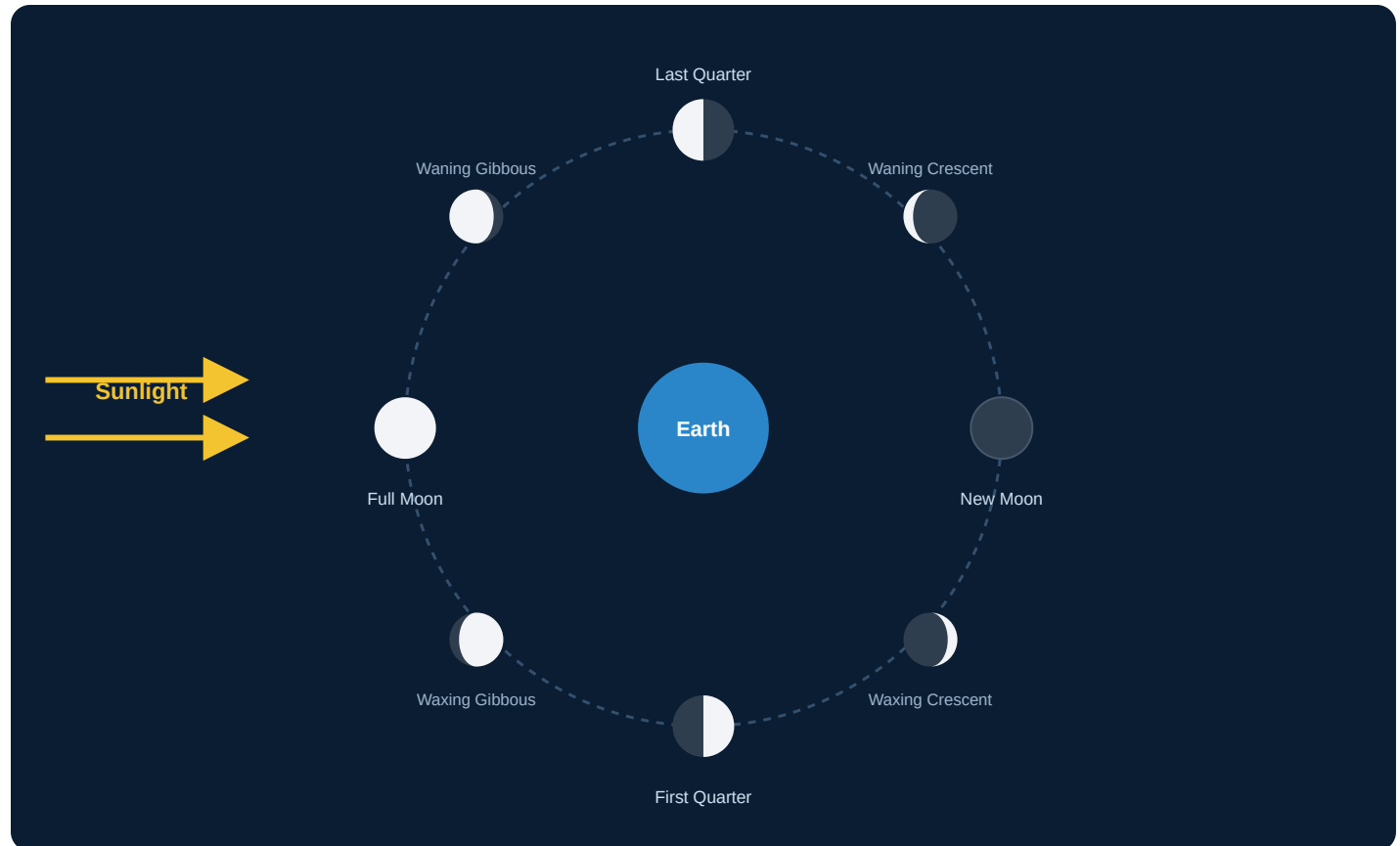


Figure 2.1 — The Moon orbits Earth. The Sun lights the same half. The labels show what we see from Earth.

Two key words:

- **Waxing** = the lit part is getting **bigger**.
- **Waning** = the lit part is getting **smaller**.

★ Easy Trick

Waxing = light is "waxing on" (growing). **Waning** = light will "wane away" (shrinking).

Fill in the blanks with **waxing** or **waning**:

- The Moon goes from new to full. The light is _____.
- The Moon goes from full back to new. The light is _____.

2.4 Solar & Lunar Eclipses

My Learning Goal

I can tell the difference between a solar eclipse and a lunar eclipse.

An **eclipse** happens when the Sun, Earth, and Moon line up in a row.

Solar Eclipse

The **Moon** is in the middle. It blocks the Sun. Order: Sun → Moon → Earth.

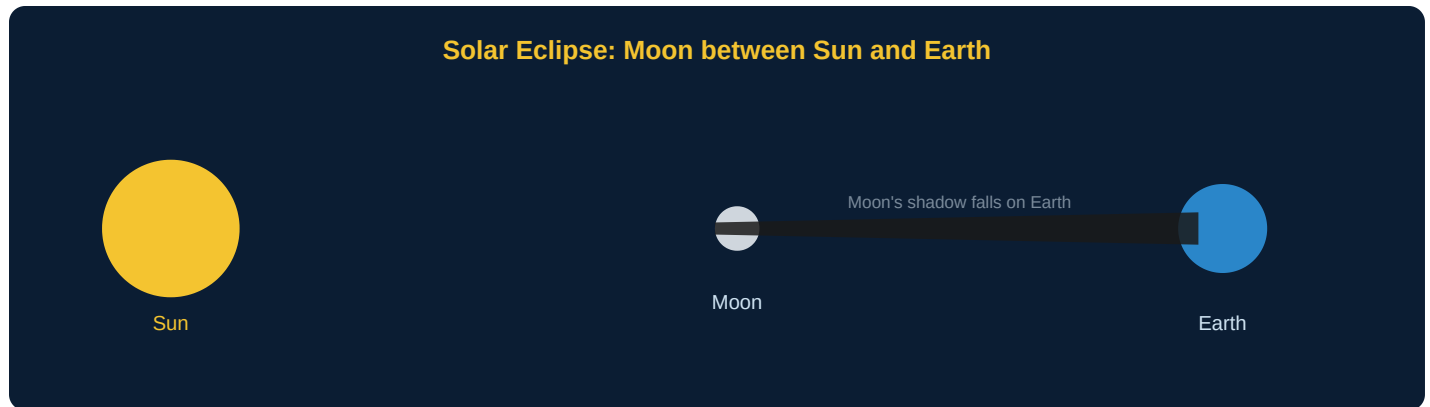


Figure 2.2 — Solar eclipse: the Moon is in the middle.

Lunar Eclipse

The **Earth** is in the middle. Its shadow falls on the Moon. Order: Sun → Earth → Moon.



Figure 2.3 — Lunar eclipse: the Earth is in the middle.

Easy Trick

Name the eclipse after what gets blocked. **Solar** = the **Sun** gets blocked. **Lunar** = the **Moon (luna)** gets darkened.

Fill in the blanks:

- In a **solar** eclipse, the _____ is in the middle.
- In a **lunar** eclipse, the _____ is in the middle.

2.5 Why We Have Seasons

My Learning Goal

I can explain that seasons are caused by Earth's tilt.

Many people think summer means Earth is closer to the Sun. **That is wrong.**

The real reason is Earth's **tilt**. Earth leans to one side, like a spinning top.

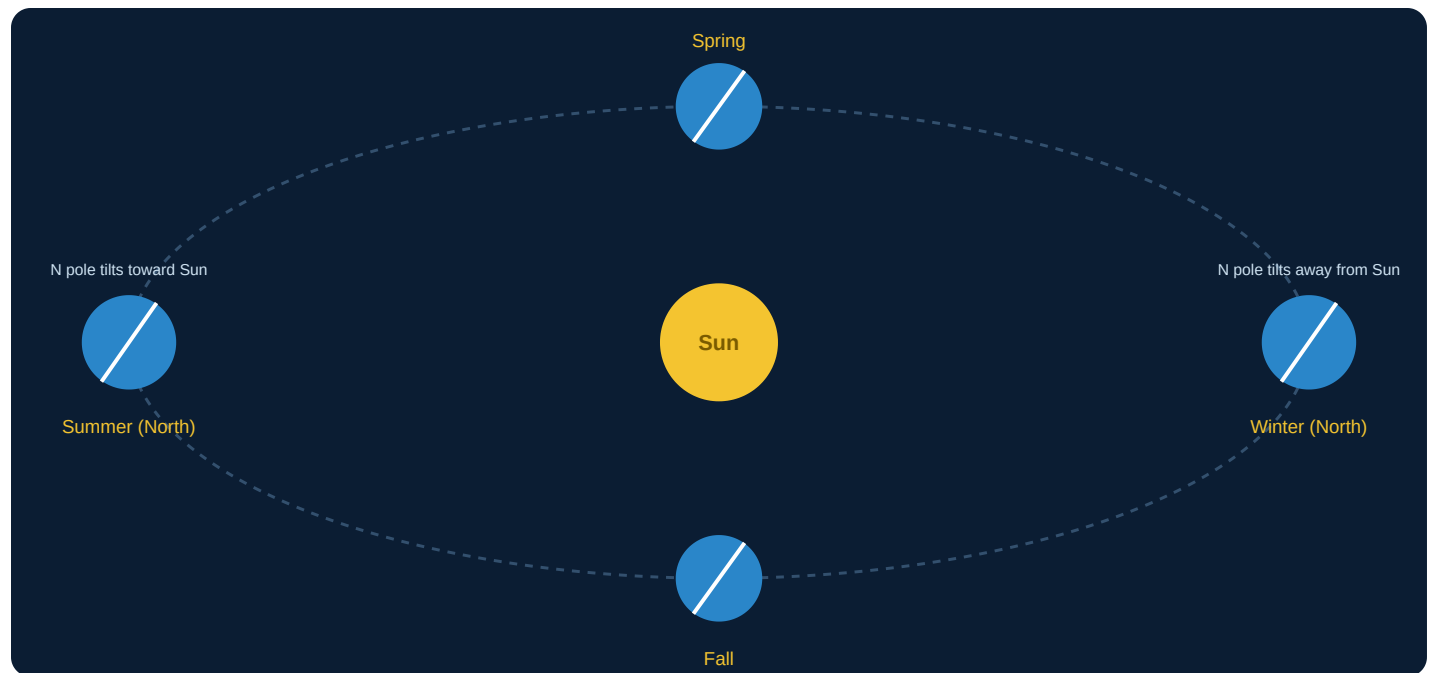


Figure 2.4 — Earth's tilt stays pointed the same way. Tilt toward the Sun = summer. Tilt away = winter.

Word Bank

toward

away

summer

winter

tilt

Fill in the blanks:

- The real cause of seasons is Earth's _____.
- When your side of Earth tilts _____ the Sun, you get summer.
- When your side tilts _____ the Sun, you get winter.

2.6 Tilt, Not Distance

My Learning Goal

I can use the two hemispheres to prove distance does not cause seasons.

Here is the proof:

Time of Year	Up North (like Georgia)	Down South (like Australia)
June	Summer	Winter
December	Winter	Summer

At the same time, the two halves of Earth have **opposite** seasons. If distance caused seasons, the whole planet would be the same. So it must be the **tilt**.

★ Surprising Fact

Earth is actually **closest** to the Sun in January, during our winter! Proof that tilt, not distance, makes seasons.

Chapter 2 Review

? Show What You Know

1. Earth spinning causes _____ and _____.
2. The Moon's light really comes from the _____.
3. "Getting bigger" light is called _____; "getting smaller" is _____.
4. In a solar eclipse, the _____ is in the middle.
5. The real cause of the seasons is Earth's _____.